

Adaptive Bayesian Restless Multi-Armed Bandit Scheduling for Electronic Warfare

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Abstract—Legacy Electronic Support (ES) receivers rely on deterministic round-robin stepping across wideband RF spectra, resulting in low intercept probabilities ($P_{\text{int}} \approx 4\%$) against modern agile radar emitters. This paper presents an online Bayesian Restless Multi-Armed Bandit (RMAB) with conjugate Beta-Bernoulli updating, hardware PLL slew penalties, and Sequential Difference Histogramming (SDIF) de-interleaving. Validated against the Alan Turing Synthetic Radar Benchmark, the proposed framework demonstrates a 70.53% mean P_{int} across 1.76 million pulses, outperforming conventional architectures by 16.4x while enforcing deterministic starvation bounds ($\tau \leq 60$ ms).

1. Mathematical Formulation (POMDP & Whittle Index)

The monitoring of $K = 16$ sub-bands is modeled as a Partially Observable Markov Decision Process. Occupancy is updated online via conjugate Beta priors $\text{Beta}(\alpha_k, \beta_k)$ without pre-training:

$$E[\theta_k] = \alpha_k / (\alpha_k + \beta_k) \mid \text{Updates: } \alpha_k \leftarrow \alpha_k + \min(h, 10) [\text{hits}], \beta_k \leftarrow \beta_k + 2.0 [\text{misses}]$$

Synthesizer tuning incurs a non-linear slewing cost proportional to log-frequency hop distance. The scheduling index balances exploitation, anti-starvation exploration, and hardware settling constraints:

$$I_k(t) = [E[\theta_k] + \tau_{\text{hop}}] \cdot \Phi_k(t) \cdot w_k + \lambda \cdot (t - \tau_k) / \tau_{\text{max}} - \kappa \cdot \log_2(1 + |k - a(t-1)|)$$

2. Multi-Split Empirical Validation (Alan Turing Dataset)

Split	Pulses	Emitters	Round-Robin P_{int}	Adaptive RMAB P_{int}	Gain
test_0.h5	29,748	78	4.54%	41.84%	9.2x
test_1.h5	4,222	7	3.96%	64.00%	16.2x
test_2.h5	792,838	19	4.37%	85.66%	19.6x
test_3.h5	16,413	32	4.37%	56.31%	12.9x
test_4.h5	104,679	44	4.39%	65.18%	14.8x
test_5.h5	368,100	58	4.38%	87.41%	20.0x
test_6.h5	453,528	28	4.37%	93.31%	21.3x
Aggregate	1,769,528	266	4.34%	70.53%	16.4x

3. Real-Time Radar De-interleaving & STANAG Interoperability

Intercepted pulse descriptor words (PDWs) feed directly into a real-time Sequential Difference Histogrammer (SDIF) that extracts fundamental Pulse Repetition Intervals (PRI), filters out harmonic ghost peaks, and correlates detected signatures against tactical emitter templates (e.g., Target Acquisition vs Fire Control radars). The resulting Electronic Order of Battle (EOB) is exportable in standardized JSON/CSV formats suitable for tactical datalinks.